Logic Minimization – Part 5

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Review

- We have discussed NAND and NOR implementations
- We will now discuss about the Exclusive OR function.

The Exclusive-OR (XOR) function

$\oplus$: Exclusive OR symbol

a	b	$a \oplus b$
0	0	0
0	1	1
1	0	1
1	1	0

Hence, $a \oplus b = ab' + a'b$

Exercise

show

$$a \oplus 0 = a$$

$$a \oplus 1 = a'$$

$$a \oplus a = 0$$

$$a \oplus a' = 1$$

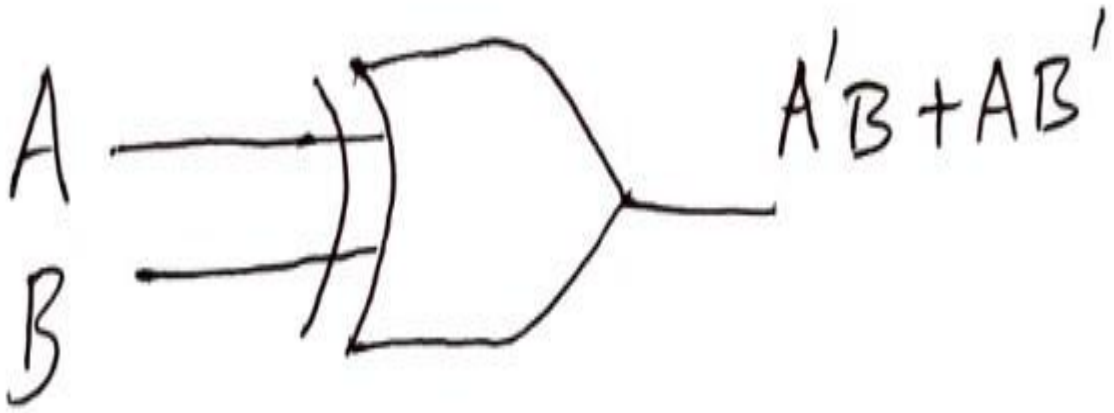
$$a \oplus b' = a' \oplus b = (a \oplus b)'$$

$$a \oplus (b \oplus c) = (a \oplus b) \oplus c$$

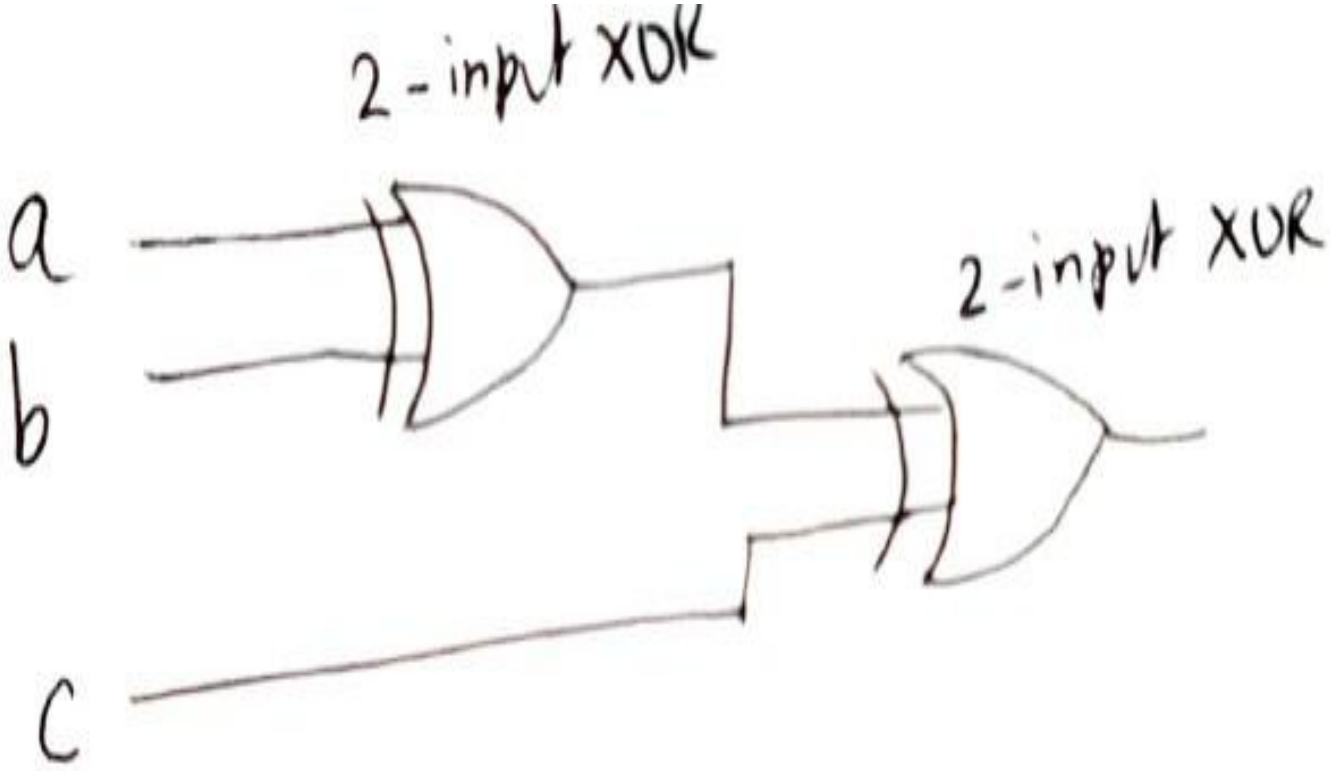
Remarks on the identities

- The associative operation given suggests XOR with three inputs. In general, multi-input XOR gates are hard to fabricate.
- Hence, 3-variable XOR may be realized via a cascade of 2-input XOR gates. We show the logic gate picture for 2-input XOR and then the cascade.

2-input XOR gate symbol



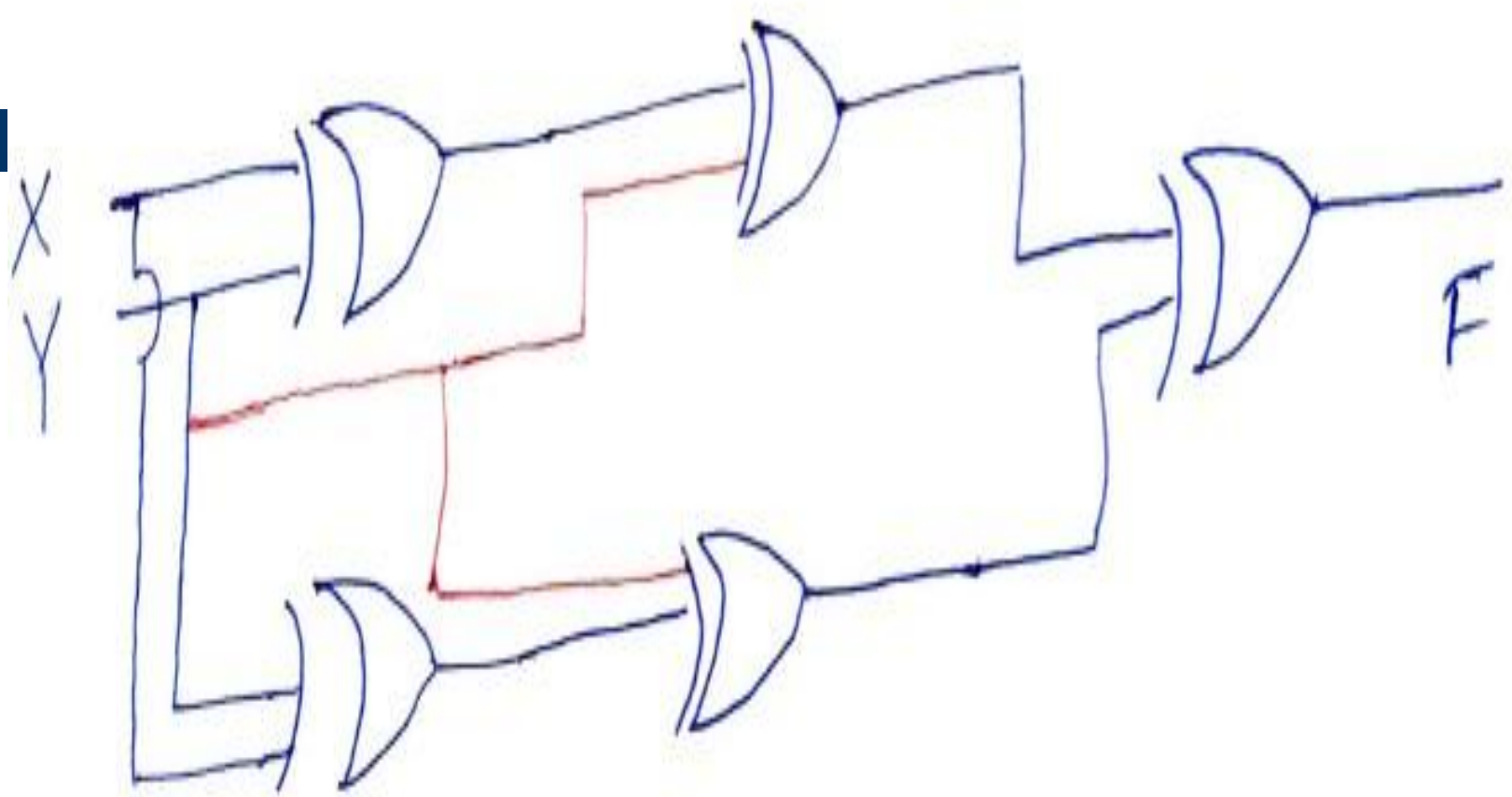
Cascading 2-input XOR gates



How to realize a 2-input XOR ?

- Generally, a 2-input XOR is also constructed with other types of gates. One can use the expression for XOR(a,b) to get a realization via AND, OR and NOT.

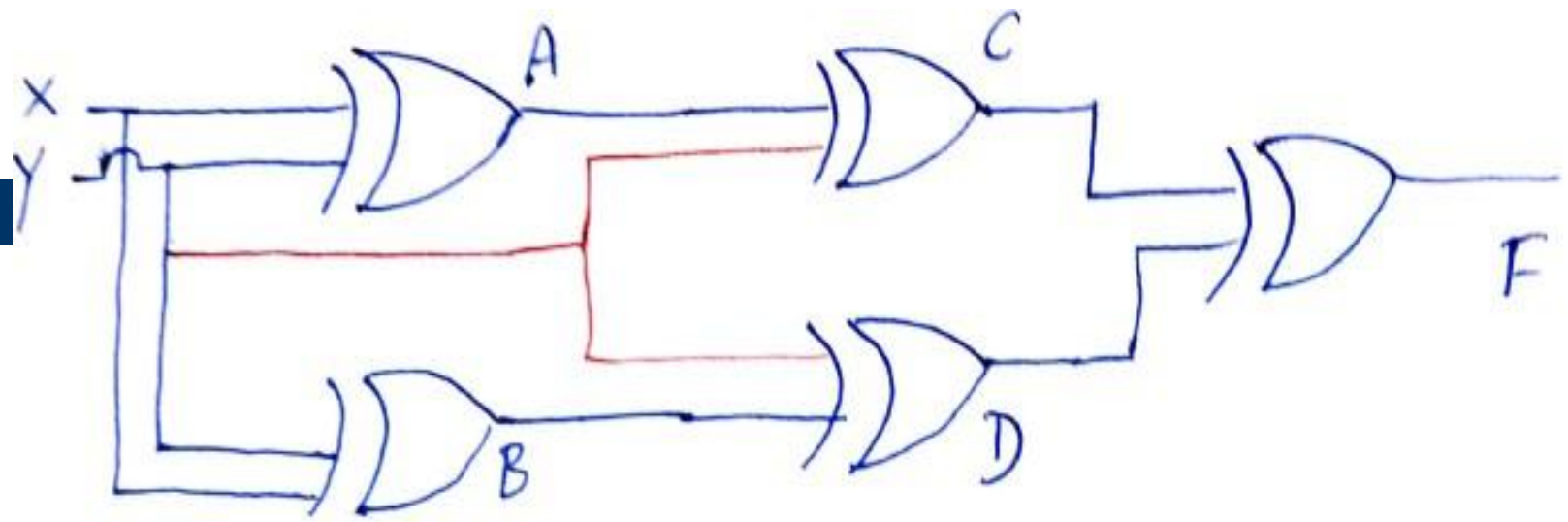
Exercise: Find the output F



Find the output F

- CLASS EXERCISE

Solution



$A = X \oplus Y$

$B = X \oplus Y$

$C = X \oplus Y \oplus Y = X$

$D = X \oplus Y \oplus Y = X$

$F = X \oplus X = 0$

The Odd function

- XOR with three variables can be converted into an ordinary Boolean function as follows.

$$A \oplus B \oplus C = (AB' + A'B)C' + (AB + A'B')C$$

$$= AB'C' + A'BC' + ABC + A'B'C$$

$$= \sum(1, 2, 4, 7)$$

More on the Odd function

- As we can observe, the 3-variable XOR is equal to 1 if one of the following is true:

  (i) Only one variable is equal to 1

  (ii) All the three variables are equal to 1

This means that an odd number of variables must be equal to 1. Hence, the 3-variable XOR is called as an Odd function.

A variation

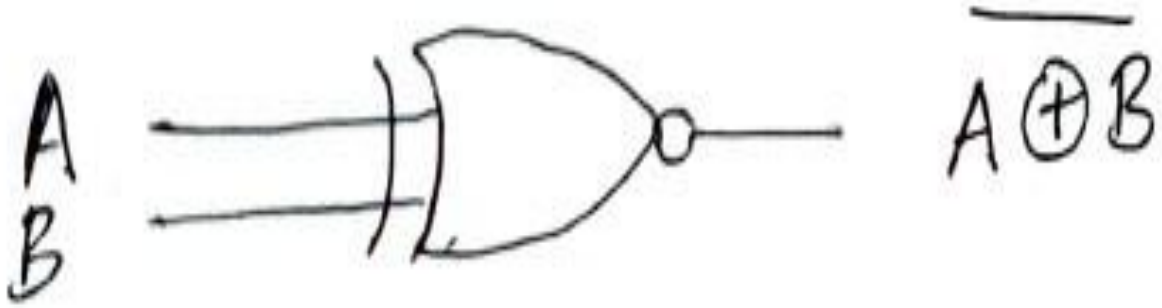
- We note that XOR (x,y) = 1 only when binary variables x and y are “different” (i.e., when

  x = 0 and y = 1 or vice-versa).
- We also have a function termed XNOR (Exclusive-NOR) which happens to be the complement of XOR. We discuss this next.

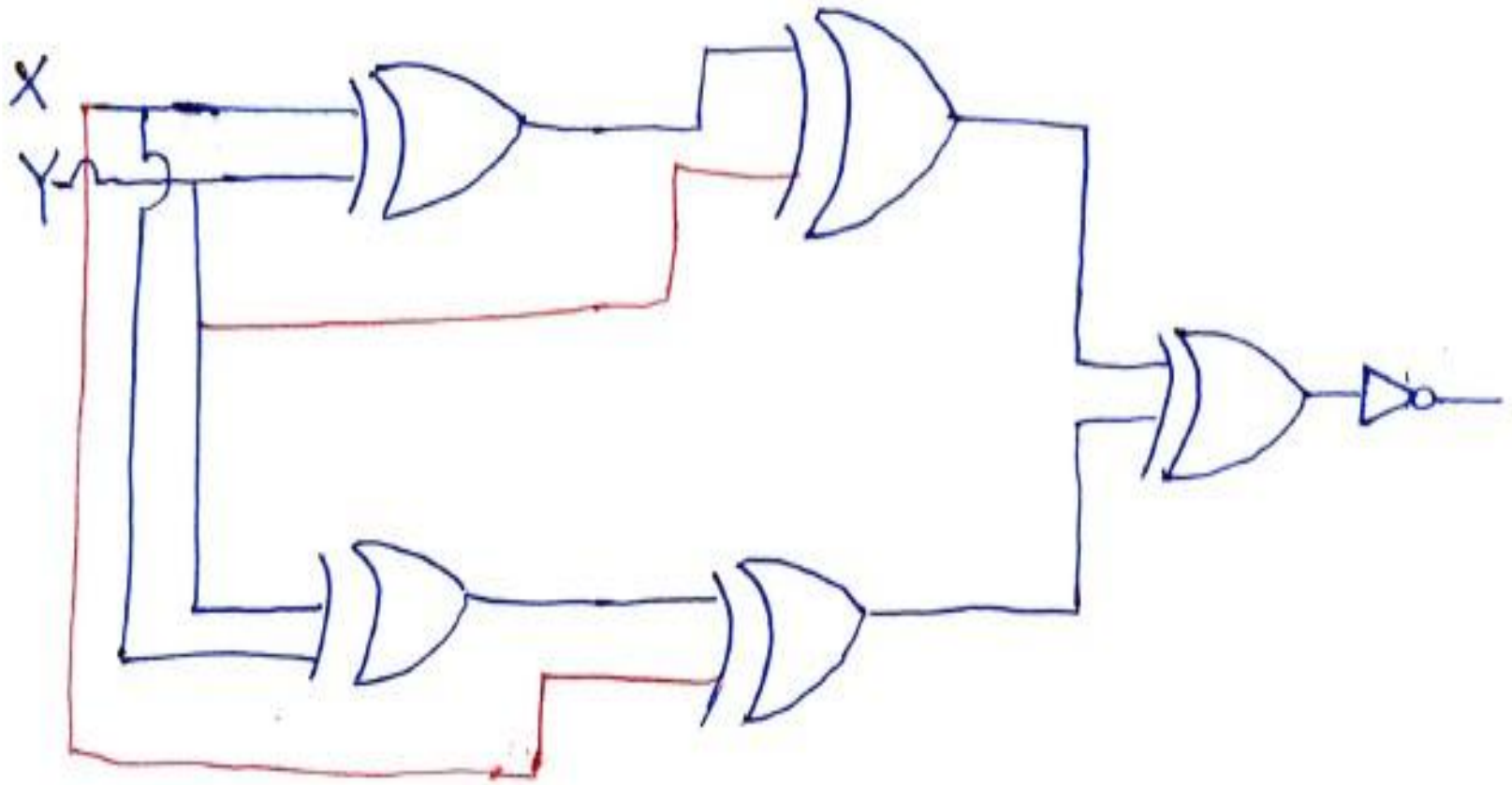
Exclusive NOR

a	b	a xnor b (a ⊙ b)
0	0	1
0	1	0
1	0	0
1	1	1

2-input XNOR symbol



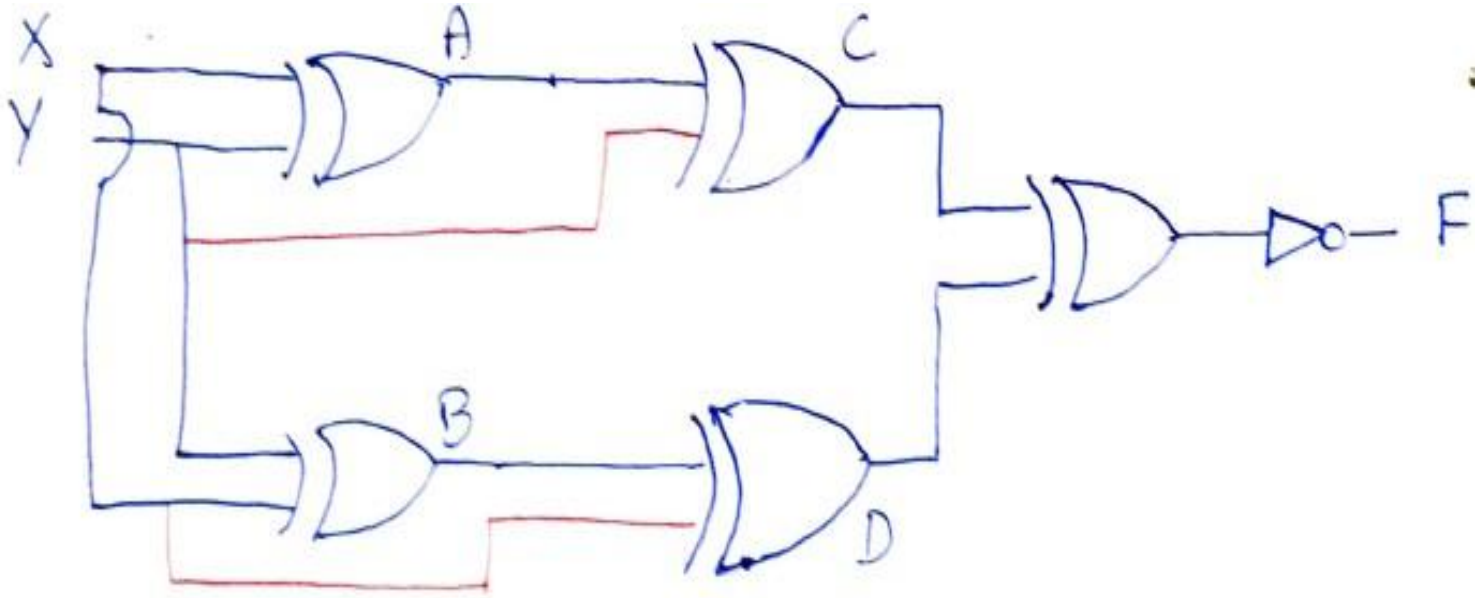
Find the output of the inverter



Find the inverter's output

- CLASS EXERCISE

Solution



$A = X \oplus Y$

$C = A \oplus Y = X \oplus Y \oplus Y = X \oplus 0 = X$

$D = B \oplus X = X \oplus Y \oplus X = Y \oplus 0 = Y$

$F = \overline{X \oplus Y} = X \odot Y$

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- C. Roth and L.L. Kinney, Fundamentals of Logic Design, Gamma Publications, 2021